CHANDIGARH UNIVERSITY

Discover. Learn Empower

Subject: Front-End-Technologies.

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Introduction:- Redux is a state management library used in modern application to manage and centralize application state. It helps maintain predictable state update using a single global store.

Here, we develop a simple counter application, using Redux. The counter starts from zero and provides 2 buttons. Increment and decrement. When the user clicks the increment button, the counter increases by 1, when the decrement button is pressed, the counter decreases by 1.

This project demonstrates the core concept of Redux such as store, action, reducers and state updates in React application.

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Code (React + Redux Toolkit)

1. Install dependencies.

```
npm install @reduxjs/toolkit react-redux
```

2. Create Redux slice → counterSlice.js.

```
import {create slice} from "@reduxjs/toolkit";
const initialState = {
    value: 0};
const counterSlice = createSlice({
  name: "counter",
  initial state,
  reducers: {
    increment: (state) => {
        state.value += 1;
    },
    decrement: (state) => {
        state.value -= 1;
}}});
export const {increment, decrement}
= counterSlice.actions;
export default counterSlice.reducer;
```

3. Create store → store.js.

```
import {configureStore} from "@reduxjs/toolkit";
import counterReducer from "./counterSlice";
const store = configureStore({
    reducer: {
      counter: counterReducer}
});  export default store;
```

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4. Wrap App with provider → index.js.

```
import React from "react";
import ReactDOM from "react-dom/client";
import App from "./App";
import {Provider} from "react-redux";
import {Store from "./store";
const root = ReactDOM.createRoot(document.
            getElementById("root"));
root.render(
  <Provider store={store}>
    <App />
  </Provider>);
```

5. Main component → App.jsx.

```
import React from "react";
import {useSelector, useDispatch} from "react-redux"
import {increment, decrement} from "./counterSlice";
function App(){
  const count = useSelector((state) => state.counter.
                                value);
  const dispatch = useDispatch();
  return (
  <div style={{textAlign: "center", marginTop: "50px
  }}>
  <h1> Redux counter App </h1>
  <h2> {count} </h2>
  <button onClick={() => dispatch(increment())}
    Increment </button>
```

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```
<button onclick={() => dispatch(decrement())}
style={{marginLeft: "10px"}}>
Decrement
</button>
</div>);
export default App;
```

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